

화학수학

열일곱 번째 수업

군론

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Determinant of a singular matrix
- Matrix eigenvalue problem

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A\text{)}$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

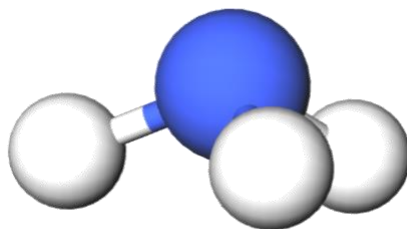
4. The **associative law** holds: $A(BC) = (AB)C$



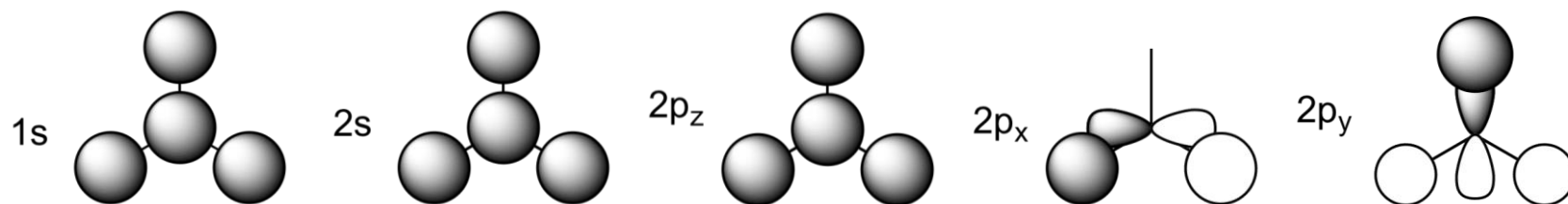
오늘의 수업

- Group theory
 - Point groups
 - Representation of a point group
 - Characters
 - Irreducible representations
 - Character tables

Group Theory and Chemistry



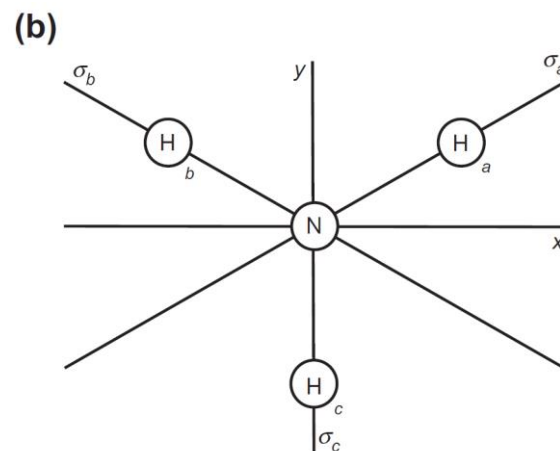
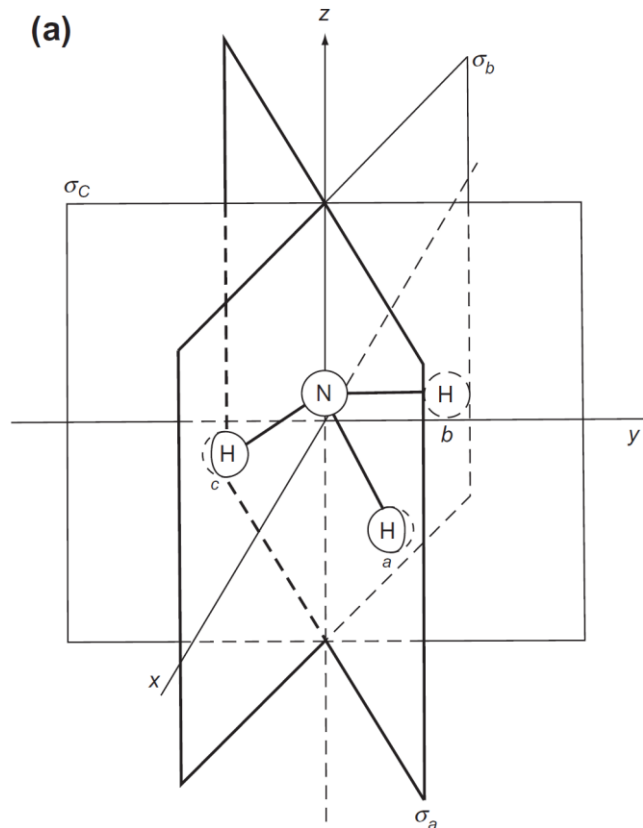
Atomic orbitals of N + 1s orbitals of H atoms



Point Group (점군)

Formed by a set of point symmetry operators
belonging to a symmetrical object

Example: Ammonia (NH₃)

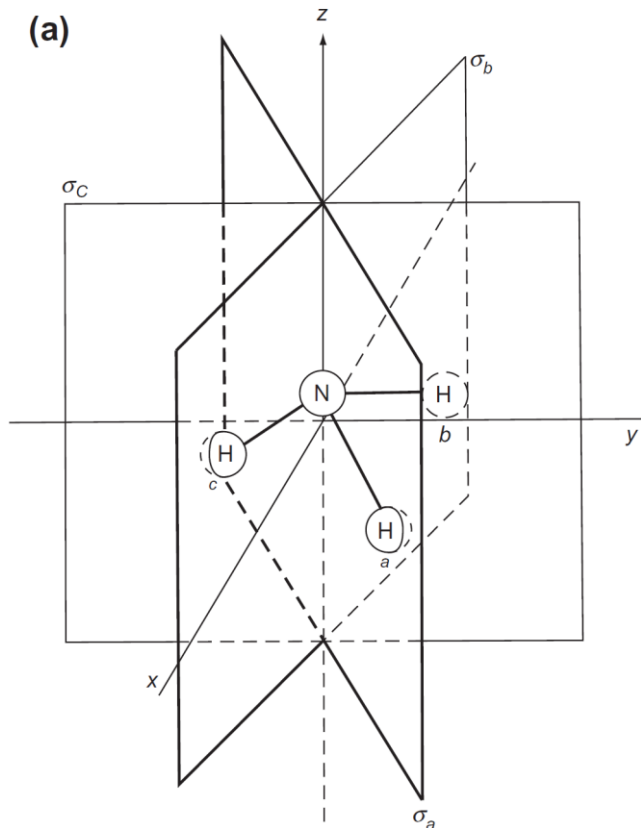


Symmetry elements: $C_3, 3 \times \sigma_v$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

Example: Ammonia (NH₃)



Symmetry elements: $C_3, 3 \times \sigma_v$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

Multiplication table:

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

Schoenflies Symbol

Each point group is assigned a symbol

Symbol	Symmetry elements
C_1	E
C_s	E, σ
C_i	E, i
C_n	E, C_n
S_n	E, S_n
C_{nv}	$E, C_n, n \times \sigma_v$
C_{nh}	E, C_n, σ_h
D_n	$E, C_n, n \times C_2$
D_{nh}	$E, C_n, n \times C_2, \sigma_h$
D_{nd}	$E, C_n, n \times C_2, n \times \sigma_v$
T_d	$E, 6 \times \sigma_d, 4 \times C_3, 3 \times S_4$
O_h	$E, 6 \times \sigma_d, 3 \times \sigma_h, 6 \times C_2, 4 \times C_3, 3 \times C_4, 3 \times S_4, 4 \times S_6$

Representation of a Point Group

Symmetry operator $\leftrightarrow$ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

Identity Symmetry Operator

$$\hat{E} \leftrightarrow \mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) ,
determine a_{ij}

Example 13.20

Find the matrix that represents $\hat{C}_{n(z)}$.

The operation transforms (x_1, y_1, z_1) to (x_2, y_2, z_2) as follows:

$$x_2 = \left(\cos \frac{2\pi}{n} \right) x_1 - \left(\sin \frac{2\pi}{n} \right) y_1$$

$$y_2 = \left(\sin \frac{2\pi}{n} \right) x_1 + \left(\cos \frac{2\pi}{n} \right) y_1$$

$$z_2 = z_1$$

Hence, the representation matrix is

$$\hat{C}_{n(z)} \leftrightarrow \begin{bmatrix} \cos(2\pi/n) & -\sin(2\pi/n) & 0 \\ \sin(2\pi/n) & \cos(2\pi/n) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Representation of the C_{3v} Group

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

Example 13.21

By matrix multiplication, show that $\mathbf{AB} = \mathbf{E}$.

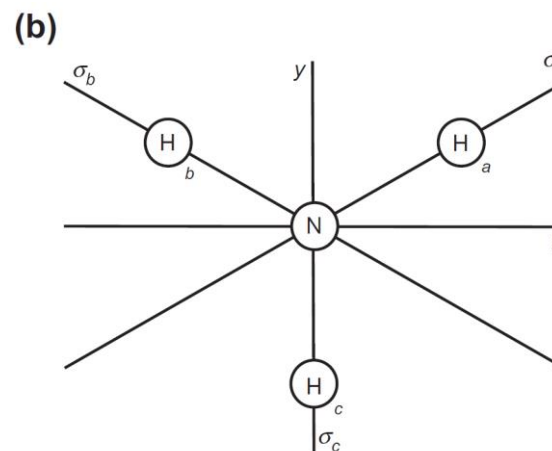
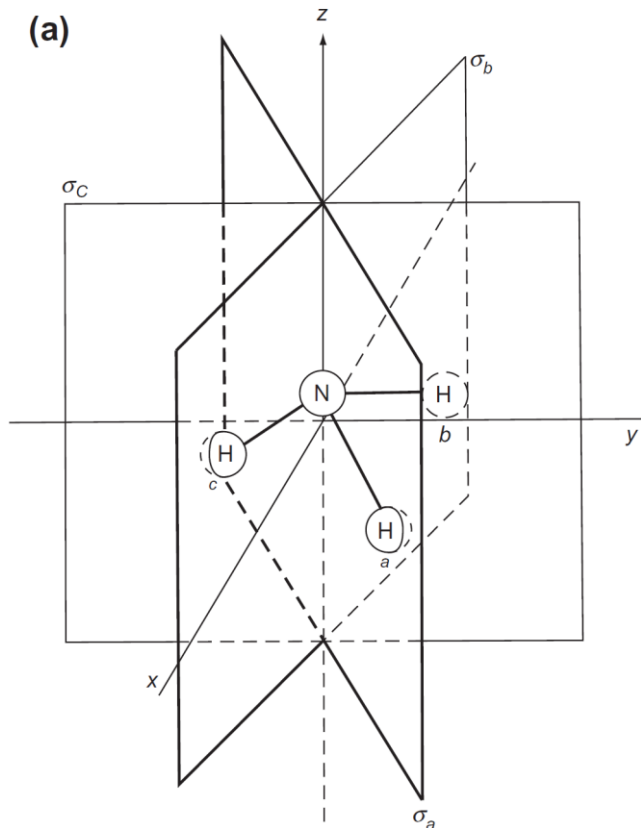
Since

$$\mathbf{A} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

matrix multiplication gives

$$\mathbf{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}$$

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

Similarity Transformation (답음 변환)

- Two representations **P** and **Q** in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when **X** is some square matrix

- Since $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \text{tr}(\mathbf{XX}^{-1}\mathbf{Q}) = \text{tr}(\mathbf{Q})$$

Character (자표)

- **Character** of a symmetry operator = **trace** of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Character of } \hat{C}_3 = \text{tr } \mathbf{A} = (-1/2) + (-1/2) + 1 = 0$$

Our Representation of C_{3v} Group

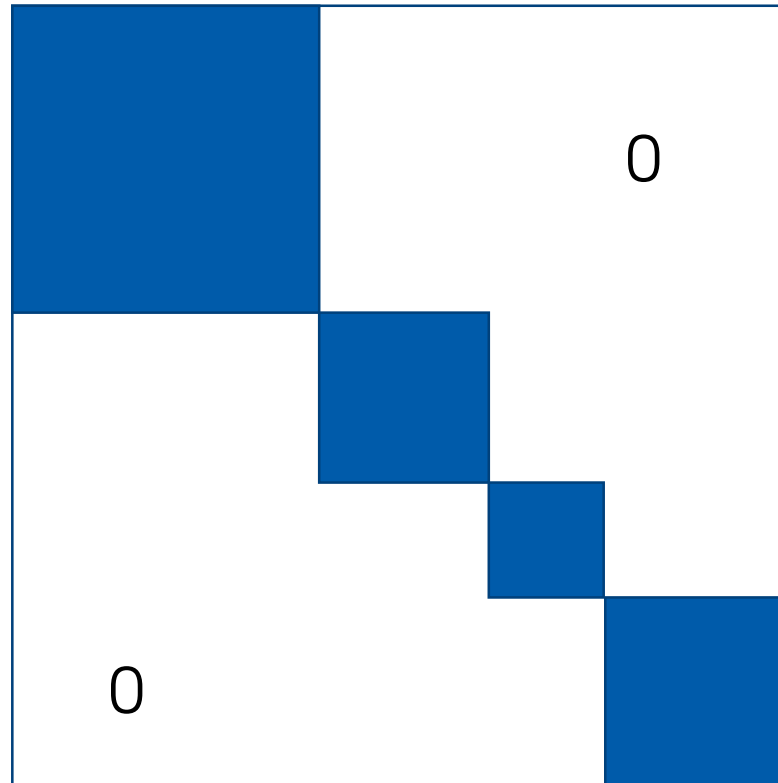
$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
3	0	0	1	1	1

Block-Diagonal Matrix (블록대각행렬)



Reducible Representation (가약 표현)

All matrices are **block-diagonal**, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{F}$$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal
by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as **irrep**

Examples of C_{3v}

- 2 by 2 blocks: one irrep

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- 1 by 1 blocks: another irrep

$$\hat{E} \leftrightarrow 1, \quad \hat{C}_3 \leftrightarrow 1, \quad \hat{C}_3^2 \leftrightarrow 1, \quad \hat{\sigma}_a \leftrightarrow 1, \quad \hat{\sigma}_b \leftrightarrow 1, \quad \hat{\sigma}_c \leftrightarrow 1$$

Characters of Irreps

Class:	E	$2C_3$		$3\sigma_v$		
Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
2 by 2	2	-1	-1	0	0	0
1 by 1	1	1	1	1	1	1

Class is constituted of symmetry operations
that are “physically identical”

Character Table (지표표)

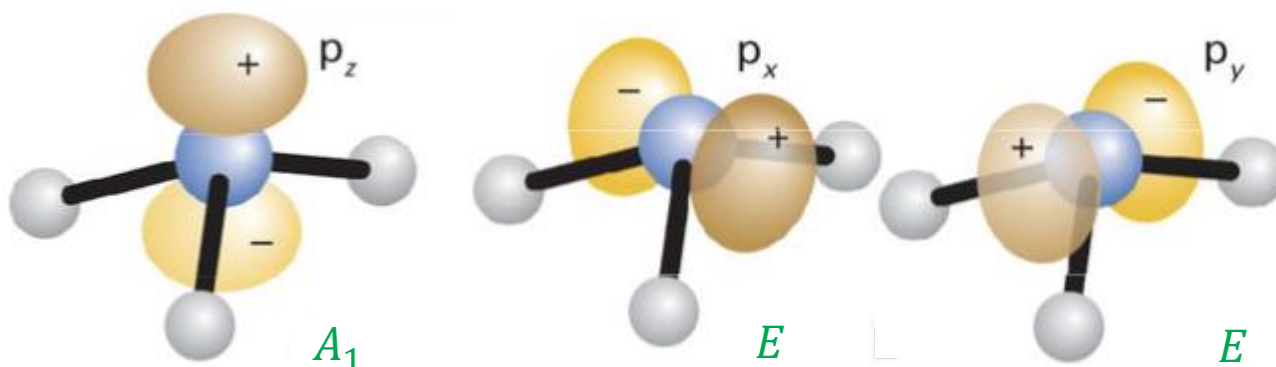
- Using group theory theorems, we can find all possible irreps and collect their characters → **character table**
- e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

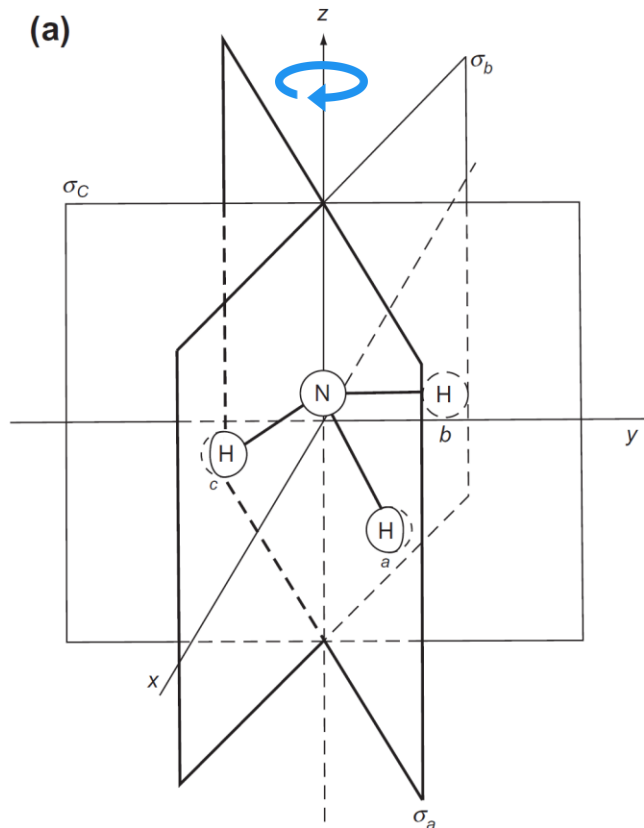
Functions and Character Table

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



(Image: https://commons.wikimedia.org/wiki/File:Orbitals_of_NH3.png)

Rotations and Character Table



C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

- $R_z \rightarrow A_2$
- $R_x \rightarrow E$
- $R_y \rightarrow E$

"Full" Character Table of C_{3v}

C_{3v}	E	$2C_3$	$3\sigma_v$	Basis components		
A_1	1	1	1	z		$x^2 + y^2, z^2$
A_2	1	1	-1		R_z	
E	2	-1	0	(x, y)	(R_x, R_y)	$(x^2 - y^2, xy)$ (xz, yz)

Application of Character Table

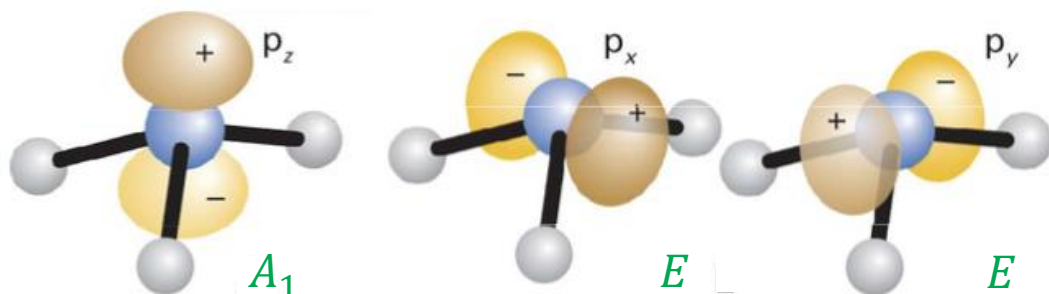
- To make a chemical bond, we need

$$\int \psi_i^* \psi_j d\tau \neq 0$$

- According to the group theory, the integral may be nonzero only if ψ_i and ψ_j belong to the same irrep of the molecular point group

Example of NH_3

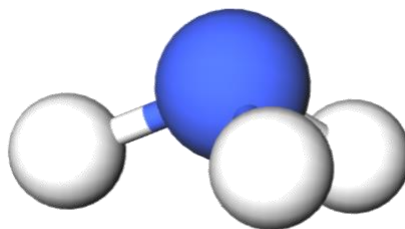
C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



H 1s orbitals



Now We Can Construct ...



Atomic orbitals of N + 1s orbitals of H atoms

